

The Application Layer: More on DNS

Lecture 7



UNIVERSITÀ DEGLI STUDI DI NAPOLI
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Application Layer

DNS

- There are two ways to identify a host: by **hostname** and by **IP address**. People prefer the more mnemonic hostname identifier, while network devices prefer fixed-length, hierarchically structured IP addresses.
 - Example: www.unina.it -> 143.225.15.50
- The **Domain Name System** (DNS) is an application-layer protocol that manages translation from hostnames to IP addresses.
- It is a **client-server protocol** in which a DNS-client asks to a DNS-server for a specific hostname-to-address translation.
- DNS **servers are often UNIX machines** running the Berkeley Internet Name Domain (BIND) software, **typically using UDP** connection and port number 53.

The Domain Name System (DNS)

- Mechanism to “convert” names to IP addresses
- Impossible to manage a centralised repository (table)
 - large number of hosts
 - geographical distance of hosts and delays
- DNS is a **distributed, decentralised system**

DNS (cont'd)

- hierarchical
- domain-based
- implemented by means of a **distributed database**

Resolver

- Program that returns the IP address associated to a name
- queries a **local DNS server**
- all messages are **UDP packets**

The DNS name space

- Naming hierarchy organised by **ICANN** (Internet Corp. for Assigning Names and Numbers)
- 250 **top-level** domains
- Each domain partitioned in sub-domains, and so on

Top-level domains

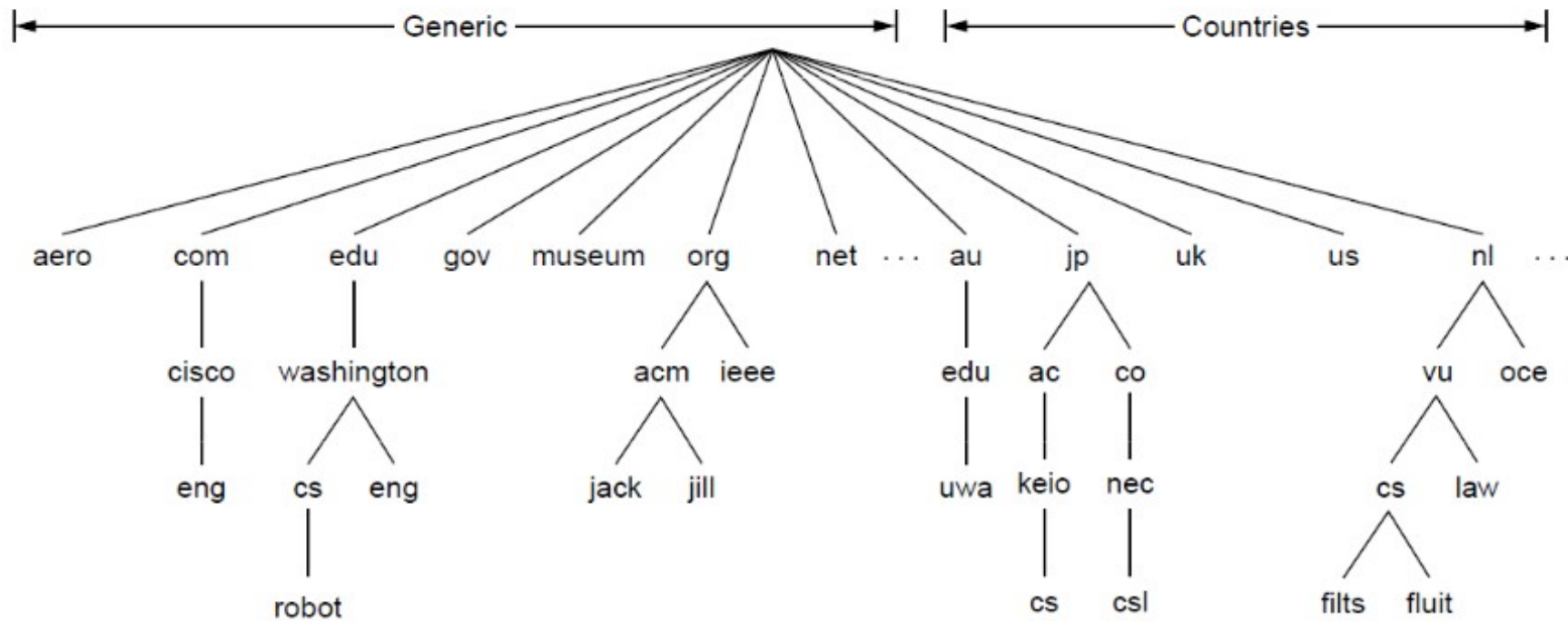
Two types

- generic (aero, com, edu, . . .)
- country (uk, uk, it, tv, . . .)

all managed by registrars appointed by ICANN

The DNS name space

- The name space is hierarchical from the root down
- Different parts delegated to different organisations



↖ The computer *robot.cs.washington.edu*

The DNS name space (cont'd)

Generic top-level domains are controlled by **ICANN** who appoints registrars to run them

Domain	Intended use	Start date	Restricted?
com	Commercial	1985	No
edu	Educational institutions	1985	Yes
gov	Government	1985	Yes
int	International organizations	1988	Yes
mil	Military	1985	Yes
net	Network providers	1985	No
org	Non-profit organizations	1985	No
aero	Air transport	2001	Yes
biz	Businesses	2001	No
coop	Cooperatives	2001	Yes
info	Informational	2002	No
museum	Museums	2002	Yes
name	People	2002	No
pro	Professionals	2002	Yes
cat	Catalan	2005	Yes
jobs	Employment	2005	Yes
mobi	Mobile devices	2005	Yes
tel	Contact details	2005	Yes
travel	Travel industry	2005	Yes
xxx	Sex industry	2010	No

Domain names

- Names are paths **upwards** from the domain to the root
 - example: `eng.mit.edu`
- **Absolute** paths end with a dot; **relative** paths are to be interpreted
- Possibility of registering under more than one top-level domain
 - example: `ibm.com`, `ibm.us`
- **cyber-squatting**: buying domains only in order to sell them later to interested businesses

Domain names (cont'd)

- Each domain controls the allocation of the domains **under** it
- The creation of a new domain requires **permission** from the parent domain
- Naming does **not** follow physical/geographical boundaries

Domain resource records

Fields of a resource record

- domain name
- time to live
- class
- type
- value

Resource record fields: Domain name (NAME)

- Tells the **domain** to which the record applies
- This field is the **main search key** for DNS queries
- Normally **several records** exist for each domain, residing at different servers

Resource record fields: Time to live (TTL)

- Indicates the **life time** of the record
 - long for stable hosts (e.g. 86, 400 for 1 day), short for volatile ones (e.g. 60 for 1 min)
 - maximum value $2^{31} - 1$, approximately 68 years

Resource record fields: Class (CLASS)

- This field is always **IN (Internet)** for Internet records
- Other codes are rarely used

Resource record fields: Type (cont'd)

Possible values:

- **SOA (Start of Authority)**: information about the server's zone (explained later), administrator's contact etc.
- **A (Address)**: 32-bit IPv4 address of some interface for the host
 - some hosts have more than one interface
 - equivalent AAAA value for IPv6 addresses
- **MX (Mail eXchange)**: gives the name of the host which accepts email in the domain
- **[see next slide]**

Resource record fields: Type (TYPE)

- **NS (Name Space)**: gives the name server for the domain; used as part of name lookup
- **CNAME (Canonical Name)**: allows for aliases; the DNS lookup proceeds with the new name
- **PTR (Pointer)**: alias used for reverse lookup: the lookup does not proceed, instead the name is returned.
- ...

Resource record fields: Type (TYPE)

Type	Meaning	Value
SOA	Start of authority	Parameters for this zone
A	IPv4 address of a host	32-Bit integer
AAAA	IPv6 address of a host	128-Bit integer
MX	Mail exchange	Priority, domain willing to accept email
NS	Name server	Name of a server for this domain
CNAME	Canonical name	Domain name
PTR	Pointer	Alias for an IP address
SPF	Sender policy framework	Text encoding of mail sending policy
SRV	Service	Host that provides it
TXT	Text	Descriptive ASCII text

Resource record fields: Value (RDATA)

- The value depends on the record type (see table in previous slide)
- Can contain a domain, a value, or an ASCII string

Domain resource records: example

; Authoritative data for cs.vu.nl

cs.vu.nl.	86400	IN	SOA	star boss (9527,7200,7200,241920,86400)
cs.vu.nl.	86400	IN	MX	1 zephyr
cs.vu.nl.	86400	IN	MX	2 top
cs.vu.nl.	86400	IN	NS	star

← Name server

star	86400	IN	A	130.37.56.205
zephyr	86400	IN	A	130.37.20.10
top	86400	IN	A	130.37.20.11
www	86400	IN	CNAME	star.cs.vu.nl
ftp	86400	IN	CNAME	zephyr.cs.vu.nl

← IP addresses
of computers

cs.vu.nl

flits	86400	IN	A	130.37.16.112
flits	86400	IN	A	192.31.231.165
flits	86400	IN	MX	1 flits
flits	86400	IN	MX	2 zephyr
flits	86400	IN	MX	3 top

rowboat		IN	A	130.37.56.201
		IN	MX	1 rowboat
		IN	MX	2 zephyr

← Mail gateways

little-sister		IN	A	130.37.62.23
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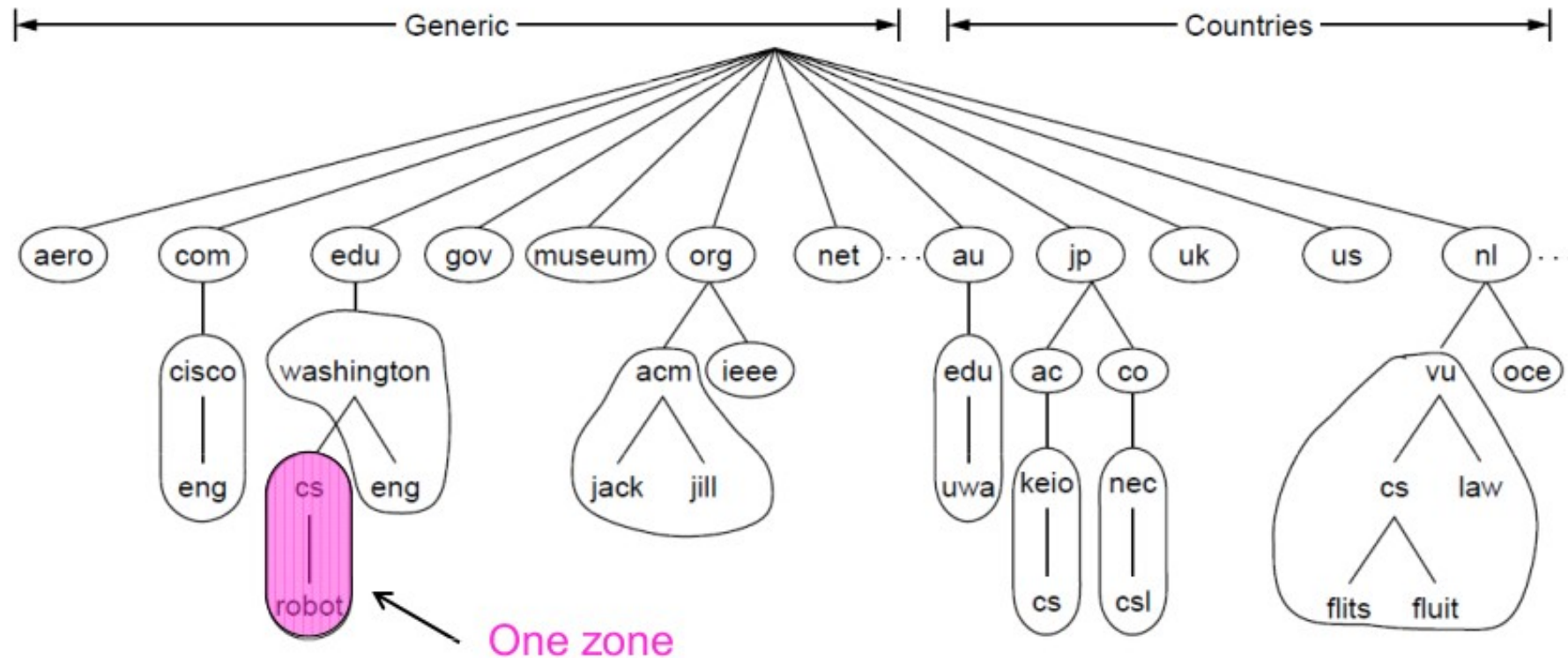
laserjet		IN	A	192.31.231.216
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Name servers

- A single, centralised DNS server makes no sense
- The DNS name space is divided in non-overlapping **zones**
- The partitioning is up to the zone administrator
- Each zone contains one or more **name servers**
 - a **primary** one and **secondary** one

Name servers (cont'd)

- Name servers contain data for portions of the name space called zones (circled)



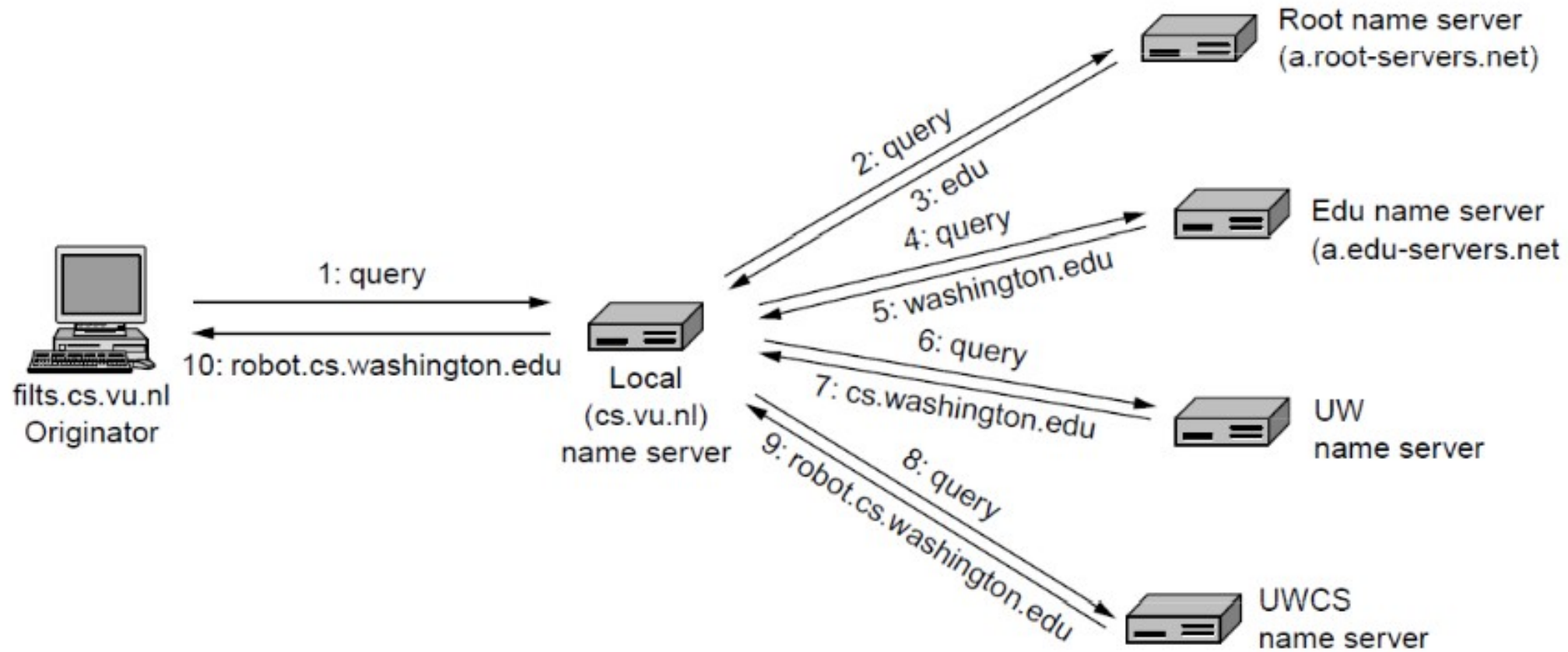
Name resolution

1. Host queries a **local** name server
2. If the domain falls under the jurisdiction (zone) of the local server, an **authoritative record** is returned
3. otherwise, the local server handles the query **recursively** and issues **remote queries**
 - if the DNS server has the address, it returns it
 - otherwise, it returns the name server for a lower zone
 - the process goes on **iteratively**, until the address is found

Name resolution (cont'd)

- these are the ones to start from when no information is available
- there are 13 of them
- a.root-servers.net to m.root-servers.net
- they are **highly replicated** servers
- they are **busy**, therefore they usually return information on lower zones (not on individual records)

Name servers: example



nslookup examples

- We can perform manually the iterative querying done by a local DNS server
- **Reverse lookup** is also possible (get name from address)

nslookup examples

The (default) server is in this case the router: 127.0.0.53 (port 53)

- The returned records are **non-authoritative** (**cached**)
- They do not come from the DNS server that manages the zone
- The local server handles the whole resolution via **iterative queries**
- The host issues a **recursive query** (either the complete answer or an error is returned)

```
andrea@turgia:~$ nslookup www.vatican.va
Server:                127.0.0.53
Address:               127.0.0.53#53

Non-authoritative answer:
www.vatican.va  canonical name = wcm-disp.spc.va.
Name:   wcm-disp.spc.va
Address: 185.152.70.33
```

Types of DNS queries

- Local DNS servers manage **recursive** queries
- service to their hosts
- Many DNS servers (the busy ones, e.g. the ones closer to the top) do **not** handle recursive queries

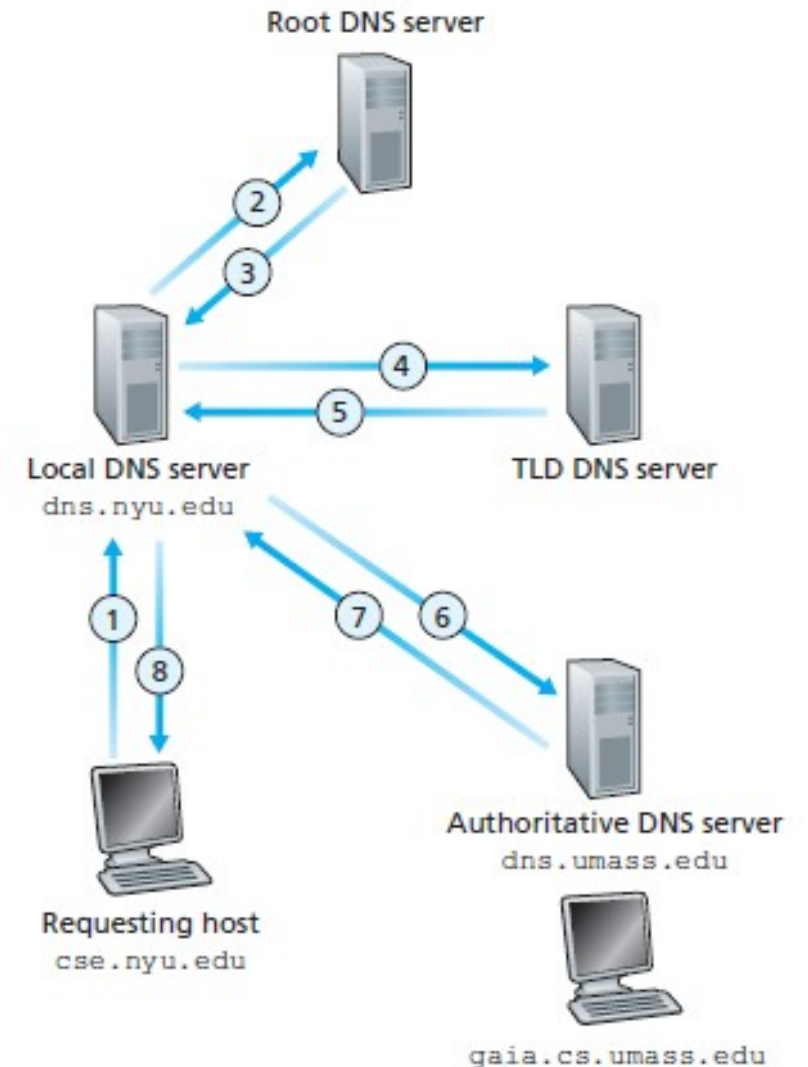
Types of DNS records

- Records are either **authoritative** or **cached**
- Cached records **expire** (recall TTL)
- Caching helps performances
- **All answers** are cached
- The address can be returned immediately by the local server if cached
- If a request for the address of **cs.mit.edu** comes and the server has the address of the DNS server for **mit.edu**, such server can be queried.
- Root servers are only queried in the absence of any information on a certain domain.

Application Layer

DNS: Resolver

- There is another important type of DNS server called the **local DNS server** (or **DNS resolver**).
- A local DNS server does not strictly belong to the hierarchy of servers and is **typically managed by ISPs**.
 - Google also provides similar servers all around the world (Google Public DNS) on address 8.8.8.8 or 8.8.4.4.
- When a host connects to an ISP, the ISP provides the host with the IP addresses of **one or more of its local DNS servers** (that are typically “close to” the host).
- When a host makes a DNS query, the query is sent to the local DNS server, which **acts as a proxy**, forwarding the query to the DNS server hierarchy.



Application Layer

DNS: Aliasing

- DNS provides an **aliasing** service for servers in which complicated/long names (aka canonical) are associated to a more simple/short ones (aka alias).
- **Host aliasing:** hosts with complicated hostnames can be **associated to more mnemonic aliases**.
 - Example: relay1.west-coast.enterprise.com -> www.enterprise.com
- Note that DNS can still be invoked by an application **to obtain the canonical (original) hostname** for a supplied alias as well as the associated IP address.

Application Layer

DNS: Load Distribution

- A DNS can be used to perform **load distribution** among replicated servers (aka, **round-robin DNS**).
- Typically, busy sites (e.g., google, amazon, CNN, etc.) are **replicated over multiple servers**, with each server running on a different end system and each having a different IP address (multiple IP addresses associated with one canonical hostname).
- The DNS database contains this set of IP addresses. When clients make a DNS query for a name mapped to a set of addresses, the server may **rotate the order in which addresses are replied**.